ECET 250E-Linear Circuit 1
Problemset 1-Chapter 1- Basic Concepts

Name: _____ Marks: _____

Chapter 1, Problem 1.

How many coulombs are represented by these amounts of electrons?

(a) 6.482×10^{17} (b) 1.24×10^{18}

(c) 2.46×10^{19} (d) 1.628×10^{20}

Chapter 1, Problem 2.

Determine the current flowing through an element if the charge flow is given by

(a) $q(t) = (3t + 8)$ mC

(b) $q(t) = (8t^2 + 4t - 2)$ C

(c) $q(t) = (3e^{-t} - 5e^{-2t})$ nC

(d) $q(t) = 10\sin(120\pi t)$ pC

(e) $q(t) = 20e^{-4t}\cos(50t)$ μC

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

Chapter 1, Problem 3.

Find the charge $q(t)$ flowing through a device if the current is:

(a) $i(t) = 3\text{A}, q(0) = 1\text{C}$

(b) $i(t) = (2t + 5)\text{mA}, q(0) = 0$

(c) $i(t) = 20\cos(10t + \pi/6)\mu\text{A}, q(0) = 2\ \mu\text{C}$

Chapter 1, Problem 4.

A current of 7.4 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 seconds.

Chapter 1, Problem 5.

Determine the total charge transferred over the time interval of $0 \le t \le 10\text{s}$ when $i(t) = \frac{1}{2}t$ A.

Chapter 1, Problem 6.

The charge entering a certain element is shown in Fig. 1.23. Find the current at:
(a) $t = 1$ ms(b) $t = 6$ ms (c) $t = 10$ ms

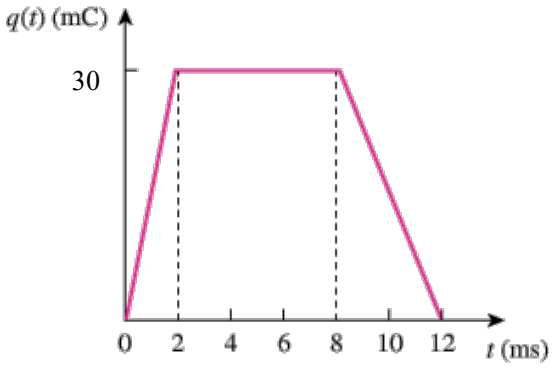


Figure 1.23
For Prob. 1.6.

Chapter 1, Problem 7.

The charge flowing in a wire is plotted in Fig. 1.24. Sketch the corresponding current.

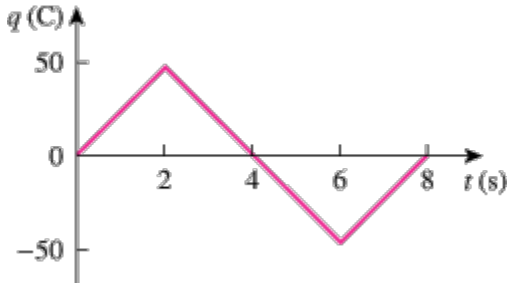


Figure 1.24
For Prob. 1.7.

Chapter 1, Problem 8.

The current flowing past a point in a device is shown in Fig. 1.25. Calculate the total charge through the point.

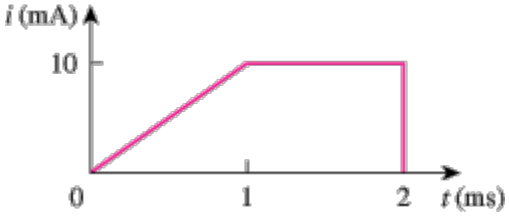


Figure 1.25
For Prob. 1.8.

Chapter 1, Problem 9.

The current through an element is shown in Fig. 1.26. Determine the total charge that passed through the element at:

(a) $t = 1$ s (b) $t = 3$ s (c) $t = 5$ s

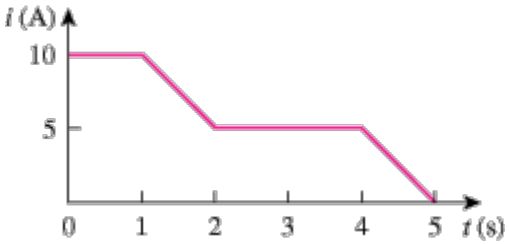


Figure 1.26
For Prob. 1.9.

Chapter 1, Problem 10.

A lightning bolt with 10 kA strikes an object for 15 μs. How much charge is deposited on the object?

Chapter 1, Problem 11.

A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h.
How much charge can it release at that rate? If its terminals voltage is 1.5 V, how much energy can the battery deliver?

Chapter 1, Problem 15.

The current entering the positive terminal of a device is $i(t) = 6e^{-2t}$ mA and the voltage across the device is $v(t) = 10di/dt$ V.

(a) Find the charge delivered to the device between t = 0 and t = 2 s.
(b) Calculate the power absorbed.
(c) Determine the energy absorbed between t = 0 and t = 3 s.

Chapter 1, Problem 17.

Figure 1.28 shows a circuit with five elements. If $p_1 = -205$ W, $p_2 = 60$ W, $p_4 = 45$ W, and $p_5 = 30$ W, calculate the power p_3 absorbed by element 3.

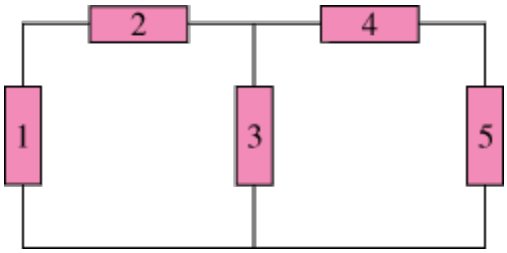


Figure 1.28
For Prob. 1.17.

Chapter 1, Problem 18.

Find the power absorbed by each of the elements in Fig. 1.29.

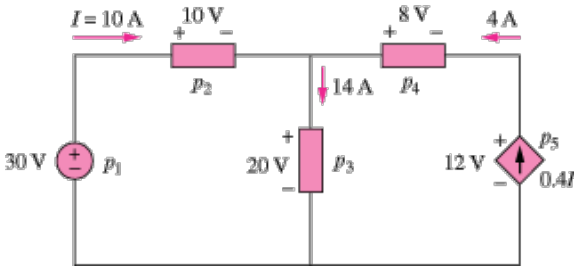


Figure 1.29
For Prob. 1.18

Chapter 1, Problem 19.

Find I and power absorbed by each element in the network of Fig. 1.30.

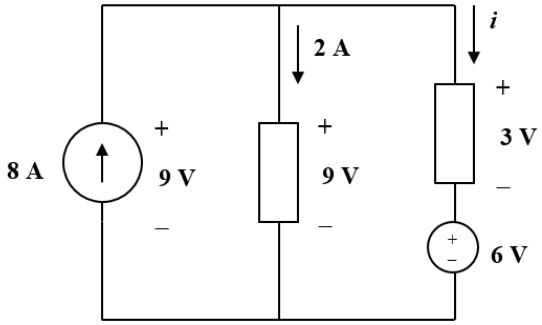


Figure 1.30
For Prob. 1.19.